DECO2300/7230 Digital Prototyping and Extended Reality
Semester 2, 2026

Tiny Worlds: An Ant-Scale XR Insect Discovery Game

Testing plan for interactive prototype 1

This project is an immersive nature-science learning experience designed to make insect education more interactive and engaging through XR. Users enter an insect world from an ant-sized perspective, select an insect from a spatial interface, and explore its anatomy through direct interaction, animation, visual feedback, and concise educational information. Prototype 1 focuses on the seven-spot ladybird and demonstrates the broader interaction flow from insect selection to anatomical exploration.

Testing Objective

The main uncertainty identified in this prototype is whether first-time users can understand the interaction language of the XR experience without extensive instruction.

The primary objective of this test is to evaluate whether users can successfully move from insect selection into ladybird exploration, discover interactive anatomical features, and understand how to interact with them using the prototype's visual and behavioural cues.

A secondary objective is to investigate whether the spatial interaction with the ladybird helps users understand the functional relationship between the elytra and the hidden flight wings, rather than simply presenting this information as text.

Testing Methodologies

Can first-time users understand and navigate the XR insect-learning interaction flow with minimal assistance, and does the interactive anatomy presentation communicate meaningful biological information?

This test will use a moderated task-based usability test combined with think-aloud observation, behavioural metrics, a short knowledge check, and a brief post-test interview.

Participants will be given goals rather than step-by-step instructions. For example, instead of being told to "point at the antenna," they will be asked to discover how the ladybird senses its environment. This allows the test to evaluate whether the prototype's visual and spatial affordances are understandable.

During the session, the moderator will observe task completion, errors, assistance required, interaction paths, and verbal comments. A short knowledge check will then evaluate whether the interactive experience communicated the intended biological concepts. Finally, Likert-scale and open-ended questions will collect subjective feedback about clarity, usability, and engagement.

This mixed-method approach is appropriate because quantitative measures reveal whether users can successfully operate the prototype, while qualitative observations and participant comments help explain why particular interactions succeed or fail.

Prototype description/requirements

The prototype is a functional Unity XR horizontal prototype representing the main interaction flow of the proposed insect-learning experience.

The testing version should contain the following working features:

Stage	Prototype feature
Insect Selection	Multiple insects arranged in a semi-surrounding spatial selection area
Hover Feedback	Pointing at an insect temporarily enlarges it
Selection	Selecting an insect keeps it enlarged and deselects the previous insect
Confirmation	Selecting the ladybird reveals its name and an Explore/Confirm button
Transition	Confirming moves the user from selection mode to ladybird exploration
Antenna Interaction	Hovering over an antenna produces movement and displays an educational information card
Elytra Interaction	Hover produces a blue rim highlight; selecting opens the elytra
Flight Wing Interaction	Wings are revealed beneath the elytra; hover produces a blue rim highlight and selection triggers wing movement
Educational UI	Contextual information cards explain anatomical structures and their functions

The prototype does not need to represent every planned insect or every final educational feature. Its purpose is to give participants a coherent impression of the overall concept and allow the core interaction language to be evaluated.

Data collection method

The moderator will use an observation sheet for every participant and record both behavioural and qualitative data.

Metric	Recording method	Purpose
Task completion	Yes/No	Measures whether the interaction flow is understandable
Wrong control attempts	Count	Tests whether controller mappings are confusing
Anatomical features discovered	Antenna / Elytra / Wing	Measures discoverability
Knowledge recall	Yes/No	Measures immediate conceptual understanding
Control clarity	1–5 rating	Measures perceived usability
Blue-highlight clarity	1–5 rating	Tests visual affordance
Information-card usefulness	1–5 rating	Tests educational feedback

Testing Setup

Before each participant:

1. Launch the prototype and confirm there are no blocking Unity errors.
2. Reset the experience to the insect-selection state.
3. Confirm the ladybird, antennae, elytra, wings, blue hover highlights, animations, information cards, and Explore button work correctly.
4. Assign the participant an anonymous identifier such as `P01`, `P02`, etc.
5. Explain that the prototype is being tested, not the participant.
6. Ask the participant to think aloud when they notice something, become confused, or decide what to interact with.
7. Do not explain where the interactive body parts are located.
8. Use the assistance scale consistently if the participant becomes stuck.

Testing process: (also considering the schedule/time)

Time	Moderator / Participant activity	Data collected
0:00–0:20	Give standard control introduction and think-aloud instruction	Initial comments/confusion
0:20–1:00	**Task 1:** “Please find and select the ladybird, then enter its exploration mode.”	Completion, time, wrong actions, assistance
1:00–3:15	**Task 2:** “Explore the ladybird and find out how it senses its surroundings and how its wings work.” Do not tell them which body parts to inspect.	Discovery order, blue-highlight use, hover/select behaviour, errors, prompts
3:15–4:00	Ask three knowledge-check questions	Learning/comprehension scores

Time	Moderator / Participant activity	Data collected
4:00–4:45	Ask quick ratings and post-test questions	Subjective ratings + comments
4:45–5:00	Thank participant, record final note, reset prototype	Researcher summary